

John Bergdall

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Employment

2025-	The University of Arkansas , Associate Professor
2022-25	The University of Arkansas , Assistant Professor
2018-22	Bryn Mawr College , Assistant Professor
2017-18	Michigan State University , Visiting Assistant Professor
2014-17	Boston University , NSF Postdoctoral Research Fellow
2013-14	Boston University , Postdoctoral Faculty Fellow

Education

2008-13	Brandeis University , Ph.D. Advisor: Joël Bellaïche
2003-08	University of Minnesota , B.S.

Other academic positions

2021-22	Max-Planck-Institut für Mathematik Visiting researcher (1 year)
2017	Max-Planck-Institut für Mathematik Visiting researcher (3 months) Institut des Hautes Études Scientifiques Visiting researcher (2 months)

National grants and fellowships

2024	National Science Foundation (DMS-2401152) “Conference: Modular forms, L -functions, and eigenvarieties”
2023-	National Science Foundation (DMS-2302284) “Collaborative Proposal: Slopes of modular forms and moduli stacks of Galois representations”
2020-25	Simons Foundation Travel Support for Mathematicians (MPS-TSM-00713782) “Eigenvarieties, automorphic forms, and Galois representations”
2014-17	National Sci. Foundation Math. Sci. Postdoctoral Res. Fellowship (DMS-1402005) “Aspects of the Langlands program via p -adic families of automorphic forms”

University awards and fellowships

2023-24	University of Arkansas Robert C. and Sandra Connor Endowed Faculty Fellowship
2022-23	University of Arkansas New Faculty Commendation for Teaching Commitment

16. [New phenomena arising from \$\mathcal{L}\$ -invariant of modular forms](#) (with Robert Pollack)
To appear in *Algebra and Number Theory*.
15. [A \$p\$ -adic adjoint \$L\$ -function and the ramification locus of Hilbert modular eigenvarieties](#) (with Baskar Balasubramanyam and Matteo Longo)
Tunisian J. Math. 7, 2025, no. 3-4, p. 515-588. DOI:[10.2140/tunis.2025.7.515](#)
14. [Huber rings and valuation spectra](#) in *Non-Archimedean Geometry and Eigenvarieties EMS Münster Lectures in Mathematics* 4, 2024, p. 1-53. DOI:[10.4171/mlm/4](#)
13. [On \$p\$ -adic \$L\$ -functions for Hilbert modular forms](#) (with David Hansen)
Mem. of the Amer. Math. Soc. 298, 2024, no. 1489, pp. iv+126. DOI:[10.1090/memo/1489](#)
12. [Reductions of 2-dimensional semi-stable representations with large \$\mathcal{L}\$ -invariant](#) (with Brandon Levin and Tong Liu)
J. Inst. Math. Jussieu 22, 2023, no. 6, p. 2619-2644. DOI:[10.1017/S1474748022000081](#)
11. [Slopes of modular forms and reducible Galois representations, an oversight in the ghost conjecture](#) (with Robert Pollack)
Proc. Amer. Math. Soc. Ser. B 9, 2022, p. 432-444. DOI:[10.1090/bproc/136](#)
10. [Reductions of some two-dimensional crystalline representations via Kisin modules](#) (with Brandon Levin)
Int. Math. Res. Not., 2022, no. 4, p. 3170-3197. DOI:[10.1093/imrn/rnaa240](#)
9. [Smoothness of definite unitary eigenvarieties at critical points](#)
J. reine angew. Math. (Crelle's J.) 759, 2020, p. 29-60. DOI:[10.1515/crelle-2017-0048](#)
8. [Upper bounds for constant slope \$p\$ -adic families of modular forms](#)
Selecta Math. 25, 2019, no. 4, Art. 59, pp. 24. DOI:[10.1007/s00029-019-0505-8](#)
7. [Slopes of modular forms and the ghost conjecture, II](#) (with Robert Pollack)
Trans. Amer. Math. Soc. 372, 2019, no. 1, p. 357-388. DOI:[10.1090/tran/7549](#)
6. [Slopes of modular forms and the ghost conjecture](#) (with Robert Pollack)
Int. Math. Res. Not. 2019, no. 4, p. 1125-1144. DOI:[10.1093/imrn/rnx141](#)
5. [An adjunction formula for the Emerton–Jacquet functor](#) (with Przemyslaw Chojecki)
Israel J. Math. 223, 2018, no. 1, p. 1-52. DOI:[10.1007/s11856-017-1611-y](#)
4. [A remark on non-integral \$p\$ -adic slopes for modular forms](#) (with Robert Pollack)
C. R. Math. Acad. Sci. Paris 355, 2017, no. 3, p. 260-262. DOI:[10.1016/j.crma.2017.01.012](#)
3. [Paraboline variation of \$p\$ -adic families of \$\(\varphi, \Gamma\)\$ -modules](#)
Compositio Math. 153, 2017, no. 1, p. 132-174. DOI:[10.1112/S0010437X16007831](#)
2. [Arithmetic properties of Fredholm series for \$p\$ -adic modular forms](#) (with Robert Pollack)
Proc. Lon. Math. Soc. 113, 2016, no. 3, p. 419–444. DOI:[10.1112/plms/pdw031](#)
1. [Ordinary modular forms and companion points on the eigencurve](#)
J. Number Theory 134, 2014, p. 226-239. DOI:[10.1016/j.jnt.2013.07.014](#)

Preprints

2. [On the variation of slopes over moduli stacks of Galois representations](#) (with Robert Pollack).
1. [On the distribution of modular points on deformation spaces](#) (link to Oberwolfach Report; with Chengyang Bao and Brandon Levin).

Computational research code

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| 2023- | Github repository: $\mathcal{L}$-invariants of modular forms (joint with Robert Pollack) |
| 2021- | Github repository: Slopes of modular forms (joint with Robert Pollack) |
| 2017- | Website: Slopes of modular forms and Fredholm series (joint with Robert Pollack) |

Invited lecture series

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|------|---|
| 2023 | Spring school on non-Archimedean geometry and eigenvarieties (Heidelberg, DE)
Four graduate-level lectures on adic spaces |
| 2022 | Graduate school on p-adic L-functions and eigenvarieties (South Bend, IN)
Four graduate-level lectures on Galois representations and eigenvarieties |

Conference and workshop presentations

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| 2026 | Algebraische Zahlentheorie (Oberwolfach, DE)
"On the distribution of modular points on deformation rings" |
| 2023 | TORA XII: Tex.–Okla. Representations and Automorphic Forms (Norman, OK)
" p -adic distributions of modular forms" |
| 2022 | PCMI: Number theory informed by computation (Park City, UT)
" p -adic distributions of modular forms"
p-adic L-functions and eigenvarieties (South Bend, IN)
" p -adic distributions of modular forms" |
| 2020 | AMS special session on automorphic forms and Galois representations (Virtual)
"Reductions of local Galois representations for eigenforms with large $\mathcal{L}$ -invariants" |
| 2019 | Moduli spaces and modularity (Oaxaca, MX)
"Explicit $\mathfrak{S}$ -modules for crystalline representations"
AMS special session on special values of L-functions and arithmetic invariants in families (Hartford, CT)
"Constant slope families of p -adic modular forms"
33rd Automorphic Forms Workshop (Pittsburgh, PA)
"Constant slope families of p -adic modular forms" |
| 2017 | AMS special session on p-aspects of arithmetic geometry (Buffalo, NY)
"Upper bounds for constant slope p -adic families of modular forms"
p-adic methods for Galois representations and modular forms (Barcelona, ES)
"Geometric properties of p -adic families of automorphic forms (and applications)" |

Conference and workshop presentations, cont.

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| 2016 | AMS special session on p-adic analysis in number theory (Minneapolis, MN)
“Some questions about slopes of modular forms”
Connecticut summer school in number theory (Storrs, CT)
“Geometric properties of p -adic families of automorphic forms and applications”
The p-adic Langlands program and related topics (Bloomington, IN)
“Slopes of modular forms and the ghost series” |
| 2015 | Boston University/Keio University joint workshop in number theory (Boston, MA)
“Slopes of modular forms and the ghost conjecture”
p-adic methods in the theory of classical automorphic forms (Montréal, CA)
“Arithmetic properties of Fredholm series” |
| 2014 | Fourth annual upstate New York number theory conference (Buffalo, NY)
“Ordinary representations on $U(3)$ and a conjecture of Breuil and Herzig” |
| 2013 | Modular forms, p-adic L-functions and Selmer groups (Oriahovitza, BG)
“Parabolizations over families of trianguline representations” |

Recent colloquia & seminar presentations († indicates colloquium or other special presentation)

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| 2026 | Inst. for Adv. Study
University of Arizona | “On the distribution of modular points on deformation rings”
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| 2025 | University of Chicago
Michigan St. University
Ohio State University
University of Arkansas | —
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| 2024 | Duke University
Oklahoma University
Oklahoma University† | “Slopes and $\mathcal{L}$ -invariants of modular forms”
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“Some distributions arising in algebra and number theory” |
| 2023 | Rice University
University of Arkansas
University of Pittsburgh†
University of Arkansas | “ p -adic slope distributions of modular forms”
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“Huber rings and valuation theory” (3 lectures) |
| 2022 | Université de Lille
Univ. du Luxembourg
Max Planck Inst.
Universität Heidelberg
Indian Inst. Sci. (IISc)
Université Laval | “Recent investigations of $\mathcal{L}$ -invariants of modular forms.”
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| 2021 | Max Planck Inst.†
Boston Univ | “Problems in the non-Archimedean theory of modular forms”
“Reductions of certain semi-stable Galois representations” |
| 2020 | University of Oregon | “Reductions of some crystalline representations” |

Conference and workshop organization

- 2024 **Co-organizer:** [Modular forms, \$L\$ -functions, and eigenvarieties](#) (Paris, France)
2015 **Co-organizer:** [Boston University/Keio University workshop 2015](#) (Boston, MA)

Colloquia and seminars organized

- 2025- **Co-organizer:** University of Arkansas Algebra Seminar
 Co-organizer: University of Arkansas Teaching and Learning seminar
2025 **Organizer:** University of Arkansas Undergraduate Math Research Symposium
2022- **Organizer:** University of Arkansas graduate learning seminars (3 semesters)
2019-20 **Co-organizer:** [Bi-College math colloquium](#)
2019-20 **Co-organizer:** [Philadelphia area number theory seminar](#)
2013-17 **Co-organizer:** [Boston University number theory seminar](#)
2014-15 **Organizer:** Boston University graduate student learning seminars
 Topics: [The local Langlands conjectures](#), [\$p\$ -adic Hodge theory](#)

Recent Teaching

- 2022- **MATH 6190V:** Topics in algebra — elliptic curves
(UArk) **MATH 3-41103:** Introduction to Abstract Algebra I, II
 MATH 30803: Linear Algebra
 MATH 32003: Number Theory

2018-20 **MATH B102:** Calculus II
(BMC) **MATH B290:** Elementary number theory
 MATH B303-4: Algebra I, II
 MATH B317: Topics in advanced mathematics: elliptic curves
 MATH B398: Senior conference: mathematics and democracy
 MATH B399: Senior conference: mathematical cryptography
 MATH B503-4: Graduate algebra I, II

Degree bearing student advising and research

- 2026 **Luke Biddle** (AB, University of Arkansas Honors College '26)
 Thesis: *Primitive Pythagorean Triples in Lean and Reduction Modulo Odd Prime Powers*
2021 **Sandy Chen** (AB, Bryn Mawr College '21)
 Thesis: *The distribution of greatest common divisor of $\mathbb{Z}[\sqrt{2}]$*
2021 **Elsa Magness** (MA, Bryn Mawr College '21)
 Thesis: *An Exposition of the Sato–Tate Conjecture for Elliptic Curves with Complex Multiplication*
2020 **Sophia Schein** (AB/MA, Bryn Mawr College '20)
 Thesis: *Hecke operators on linear representations over finite fields*
2016 **Alexander Peraire-Bueno** (Boston University Academy thesis)
 Thesis: *Counting with partitions*

Other student research, mentorship, and service

- 2025- **Ph.D. Thesis Committees (University of Arkansas)**
Total number of students: 1.
- 2024- **Honors College Thesis Committees (University of Arkansas)**
Total number of students: 9.
- 2019 **Summer Science Research (Bryn Mawr College)**
Students: Sandy Chen and Sophia Schein
PROMYS research project writer
Students: Eric Tang, Aryaman Srikant, Emily Huang, and Aidan Griffin
Title: *Representation theory and Dickson's theorem*
- 2016 **PROMYS research project mentor**
Students: David Amirault, Vanshika Jain, Roshan Padaki, and Sabir Shaik
Title: *Slopes of Newton polygons*

College and university service

- 2026- **Promotion and Tenure Committee** (Univ. Ark. Math.)
- 2025 **Fulbright Program Campus Committee Member** (Univ. Ark.)
- 2023- **Curriculum committee** (Univ. Ark. Math.)
- 2023-25 **Ad-hoc graduate qualifying exam committee** (Univ. Ark. Math.)
- 2020 **Graduate Council** (Bryn Mawr Coll.)
Director of Graduate Studies in Mathematics (Bryn Mawr Coll., Math.)
- 2019-20 **Committee on Undergraduate Awards and Fellowships** (Bryn Mawr Coll.)

Non-college and -university professional service

- 2024- **Peer review for grants and other proposals**
National Science Foundation, Banff International Research Station
- 2016- **Peer review of publications**
Journals: 15 total including Algebra and Number Theory, Duke Math. J. Invent. Math.
- 2023 **TORA XII Graduate Student Q & A session**
Facilitated question and answer session for graduate student conference participants
- 2020 **Panel: Cross Atlantic representation theory and other topics online**
Topic: "How individuals are dealing with the pandemic"
- 2019 **Panel: Philadelphia Undergraduate Mathematics Conference**
Topic: Professional development
- 2016 **Panel: Boston College mathematics graduate student association**
Topic: Professional development

Substantial professional development

- 2024 **Inspiring Inquiry and Preparing Lifelong Learners** (Certificate provided by the Association of College and University Educators (ACUE))
- 2020 **Teaching and Learning Institute partnership**
Partners: Sara Grossman, Kirtee Ramo, and Yeipyeng Kwa
Online Teaching Institute (Bryn Mawr College)
- 2019 **Cornell Interactive Theater Ensemble workshop**
Theme: "Hang in There and Be Tough"
Posse Plus Retreat
Theme: "The State of our Union"
- 2018-19 **Teaching and Learning Institute partnership**
Partner: Jake Ogata Bernstein

Professional membership

- 2020-23 **Mathematical Association of America**
- 2008-24 **American Mathematical Society** (gap 2014-15)